

The interconnection between water quality level and health status: An analysis of *Escherichia Coli* contamination and drinking water from Nepal

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ABSTRACT

Health outcomes can be remarkably dependent on the quality of drinking water. This study employs primary survey data collected in May 10 – July 27, 2016 to investigate the effects of water quality, measured by the presence of the *Escherichia Coli* (*E. Coli*) bacteria, on the health outcomes of households in Nepal. The health outcomes are based on the self-reported health status as well as the actual health condition of the members of the household. The novel contribution of this paper is that we investigate the health impacts of water quality by considering water quality itself as an intermediary process that could be affected by factors like household's cleanliness behaviors and hygienic infrastructures. Using a control function approach to account for potential endogeneity issues, the results indicate a significant effect of *E. coli* on the health status of individuals. Findings indicate the difference in expected diarrheal incidence to be about 0.913 units higher when *E. coli* is present in households' drinking water. The impact of *E. coli* is evident in both the self-reported health status and the actual health outcome, and the results hold across various empirical models and specifications. From a policy perspective, our results indicate the pressing need for policymakers to implement water quality tracking programs, and also to raise awareness among households regarding adequate cleanliness practices and hygiene infrastructures.

1. Introduction

Drinking water can significantly affect health outcomes because of its direct use on our daily life. The adverse impacts on the health outcomes are perhaps more acute in developing nations, as majority of the households there tend to store water in potable containers. This behavior is driven by factors such as irregular water supply and high level of water pollution in these countries, which compels the households to stockpile drinking water for later use. However, inadequate practices of cleanliness behavior and lack of hygienic infrastructure, particularly during transporting or when storing the water, have been found to be a frequent source of contamination of households' drinking water [1–3]. Poor water quality will, in turn, have a negative impact on the health outcome of people. For

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instance, a report published by the World Health Organization [4] indicated that improved water supply alone could reduce diarrhea-related morbidity incidences by 6–25%.

While the negative relationship between drinking water quality and health outcomes have been documented and well understood, the health implications of poor water quality are not directly observable and can potentially occur via intermediary processes. One intermediary process could be the level of microbial contamination in the drinking water, that can transpire as a result of households' poor cleanliness behavior and inadequate hygienic infrastructures. This study employs primary survey data collected from Siddharthanagar, Nepal in 2016 to explore the relationship between households' health outcomes, cleanliness behavior, and hygienic infrastructures by introducing an intermediary process: the level of contaminants in the households' drinking water. Nepal is a developing nation situated in South Asia, with a rapidly increasing population of 28 million people [5]. The water quality concerns in Nepal stem from inadequate access to safe drinking water to the surging population in the country. It is estimated that only 27% of the population have access to safely managed water supply free from contamination in Nepal [6].

In addition to improved water sources, another source of concern stems from the lack of access to proper sanitation facilities. According to UNICEF [7], over 33% of the total population in Nepal do not have access to improved sanitation, which further elevates the chances of microbiological contamination in drinking water. The extent of water pollution and waterborne diseases are also propelled by a lack of cleanliness behavior practiced at the household level. While handwashing is considered a crucial cleanliness practice, only 40% of the population in Nepal have access to handwashing facility with soap and water at home [6]. These existing shortcomings have collectively contributed to poor water quality, and almost 82% of households in Nepal are considered to be at risk of microbial contamination in drinking water [7].

An inevitable outcome of microbial contamination in drinking water is the prevalence of waterborne diseases like diarrhea, which is evidently high in Nepal. The country faces a soaring rate of diarrhea-related morbidity incidences, and about 5.91% of deaths in the country were attributed to diarrheal incidences in 2017 [8]. Poor health outcomes will also reduce labor productivity and ultimately create a significant toll on the economy of the country. The economic burden of poor health stemming from water quality issues could be devastating to developing nations like Nepal, where the per capita GDP is USD 1043.11 [9]. Our study area, Siddharthanagar, is considered as a gateway to Lumbini, the birthplace of Buddhism. This region sees a large influx of tourists that come to visit Lumbini every year, and the local economy of Siddharthanagar is highly dependent on the tourism sector. In a study by Murphy et al. [10] on 'Travelers Diarrhea', they find a significant impact of contaminants in water on the prevalence of diarrhea among tourists in Nepal.² Poor water quality and the associated health concerns can, thus, also be detrimental to the economy in the Siddharthanagar region.

As such, there is a dire need for effective water handling policies to be implemented in Siddharthanagar and Nepal in general, which can only be possible if relevant studies on water handling behaviors are carried out. This study employs a conceptual framework of health production function to analyze the impact of drinking water quality on health outcomes among households in Siddharthanagar, Nepal. Studies on water handling behavior and health outcomes are not new, and there are a number of studies centered around this topic. However, most past studies on water quality behaviors have either only investigated the impact of water quality on health outcomes [11–15], or studied the effect of cleanliness behavior and hygienic infrastructures on health outcomes [16–20]. The novel contribution of this study is that we deviate from these past studies and instead focus on the effect of water quality on health outcomes by considering water quality level itself as an intermediary process that could be affected by households' cleanliness behavior and hygienic infrastructures.

The water quality level in our study is based on the level of microbial contamination measured in terms of the presence of total coliform bacteria in the drinking water. Total coliform bacteria are a group of easily cultured organisms and are frequently used to indicate the level of water quality [21]. Coliforms are widely used to indicate water quality levels as they are easy to isolate, are present in larger numbers, and they usually survive longer in an aquatic environment than viruses, parasites, and other types of bacteria. *Escherichia Coli* (*E. coli*) is a species of fecal coliform bacteria that has a direct link to water quality [22], and it is used as a measure of microbial contamination of water quality level in this study. The level of *E. coli*³ in the households' drinking water is scientifically tested, with the presence of the *E. coli* bacteria indicating that water has potentially been exposed to feces, and immediate risk to human health exists [23].

Drinking water quality could be affected by a myriad of factors, two such being the household's cleanliness behavior and hygiene infrastructures. Consequently, a direct estimation of the effect of *E. coli* on health outcomes could suffer from potential endogeneity issues due to omitted variable bias. To circumvent endogeneity concerns, this study employs a control function (CF) [24] approach to control for possible endogeneity of *E. coli* in the drinking water. The results from the ordered probit model indicates a significant interrelation between the *E. coli* in drinking water and households' self-reported health status. Results suggest that households with the presence of *E. coli* bacteria in their water were more likely to report poor health status. This finding was consistent under self-reported health status and also when we examined the actual health outcomes of the individuals. The negative binomial estimates show that the difference in the expected incidences of diarrhea to be 0.913 units higher when *E. coli* is present in drinking water. Furthermore, the results remain robust under various model specifications, and even when the sample is investigated separately by looking only at the children population, the adult population, and when we focused only on the urban sample. From a policy perspective, the results in this

² Since tourists often stay in hotels, the concept of "travelers' diarrhea" might not seem relevant in our study where we are investigating the impact of hygiene infrastructure and cleanliness behaviors on *E. coli* level. However, the transmission of diarrhea (from *E. coli*) to tourists is equally likely from the hotels if the staffs in these hotels do not follow adequate hygiene and cleanliness practices.

³ Most of the *E. coli* is harmless and does not cause any harm. In this research, we refer to *E. coli* 0157:H7 which causes the most outbreaks and health hazard.

study suggest a need to implement water quality tracking programs; to increase investment in raising awareness regarding cleanliness practices via social media campaigns and school-level curriculums; and to subsidize costs of acquiring hygienic infrastructures like flush toilet and tube well systems, all of which could drastically lower water-related concerns in the Siddharthanagar municipality.

2. Literature review

Self-reported health has been widely used as an indicator of health outcomes in primary survey studies [25]. The reported health behavior of an individual, however, can be influenced by numerous factors they are exposed to, such as environmental amenities and socioeconomic status [26–28]. Past literature on health behavior have indicated a substantial and significant relationship between self-reported health and water quality. For instance, Kile et al. [29] found a positive relationship between arsenic exposure in drinking water and self-reported nausea during pregnancy among a cohort of pregnant women in Bangladesh. Napier et al. [30] investigated the association between fecal indicator bacteria and self-reported illness among swimmers in the United States (U.S.) beaches and reported a similar positive relationship between the two. However, one caveat with these studies is that they assume self-reported health status as an exogenous variable. Self-reported health, however, could be influenced by multiple factors varying at the individual level, which creates a bias in the reporting if not properly accounted for [31].

This paper explores the interconnection between households' hygienic behavior and infrastructures, health outcomes, and microbial contamination in their drinking water. The health outcome is measured in terms of the self-reported health status and also on the frequency of diarrhea occurrences in the household. We incorporate the occurrence of diarrhea as an additional indicator of health status to overcome the issue of potential bias resulting from employing self-reported health status. In regards to the relationship between hygienic behavior, infrastructures, and health outcomes, there are several studies that have highlighted the direct effect of factors such as improved hygienic practices and infrastructure on diarrheal incidences. Among the various possible cleanliness practices, hand washing has been found to significantly reduce the incidence of diarrheal diseases [16–20].

Likewise, hygiene infrastructure within and outside the household is another factor that could have a considerable effect on health outcomes. A number of studies have attributed households' hygienic behavior to health outcomes [20,32,33]. Trinies et al. [34] found that in addition to hygienic behaviors inside the household; school-based water, sanitation, and hygiene (WASH) interventions could also reduce diarrhea and respiratory infection among students. Access to the safe water source is another crucial factor that has been attributed to reducing diarrheal incidences [35,36]. Similarly, other factors such as the gender and age of the child, the level of education of the mother, whether the mother has had any immunization, and the type of living area of the household have all been found to affect diarrheal incidence rates [37].

The prevalence of diarrheal incidence is frequently a direct consequence of microbial contamination in drinking water. The contamination of drinking water could result from a multitude of factors, such as poor water handling behavior, the source of drinking water, or the cleanliness behavior of the household [1,3,38,39]. In regards to the cleanliness behavior of the households, the process of handwashing and the type of handwashing water are also significantly associated with *E. coli* contamination [40,41]. Daneshmand et al. [40] identified that animal ownership also increases *E. coli* contamination in soil, and *E. coli* in the soil are correlated with contamination on hands before washing. Finally, the duration of storage as well as the material of the storage vessel can both also affect the quality of the drinking water [42,43].

The scope of our study fits in with the recent literature on health outcomes and hygienic behavior as we demonstrate a complete scenario of the system discussed above. We investigate the effect of hygienic behavior and infrastructures on health outcomes by exploring the intermediary process of microbial contamination in the drinking water. The contributions of our study are on several fronts. Waterborne diseases like diarrhea have had a significant toll on the health of the labor force and the economy in Nepal; yet, there are very few studies that have addressed the association between hygiene behavior, contamination, and health problems. Kunwar & Bohara [44] investigated whether the deviation between households' perception of their water quality and the objective water quality level could affect their water handling behaviors in Siddharthanagar, Nepal. While their study highlights the transmission route between household behavior and water contamination, the impact on health outcomes itself is excluded. Acharaya et al. [45] find poor water handling practices and partial breastfeeding to be the major risk factors towards the burden of diarrhea in Nepal. Karki et al. [46] identified factors such as knowledge and practice, education, malnutrition, and floods to be an important determinant of diarrhea in mid-western Nepal. The latter two studies highlight the direct relationship between health outcomes and behaviors; however, the intermediary process (i.e., microbial contamination) is ignored.

This study is one of the few to investigate the relationship between hygiene behavior and health outcomes by introducing an intermediary process and is the only study in the context of Nepal. Second, our paper relies on scientifically gathered data on water quality to measure the intermediary process, thereby presenting a more robust and methodical approach to incorporate this information in the health outcomes literature. Third, we employ a number of econometric techniques to analyze the data, which makes our findings and the resulting policy implications more robust. Overall, this study provides evidence that better hygienic infrastructure and cleanliness behaviors reduce microbial contamination and increases water quality level, which ultimately mitigates potential health

risks.

3. Data collection method

This study uses primary survey data that was collected to explore the water handling behavior of households from the Rupandehi district in the Terai region of Nepal. The survey was conducted in the urban city of Siddharthanagar and two its adjacent rural settlements, Basantapur and Bagaha VDC's.⁴ The survey took place between May 10 – July 27, 2016 and was done using in-person interviews.⁵ The survey protocol included expert interviews with water management officials, focus-group discussions, debriefing sessions, a pilot survey, and the final survey. The households were selected using stratified random sampling and were interviewed to assess their knowledge, attitudes, and practices regarding water use and hygiene behaviors in the urban and rural communities. In addition to the in-person household interviews, a water quality test was carried out on 314 households to measure the presence of the *E. coli* bacteria on the household's drinking water. The water sample collected from the households were taken to a laboratory to carry out the testing of *E. coli* bacteria at the end of each day after completing the survey interviews.

The test to detect the *E. coli* bacteria was conducted using a single-use disposable testing kit developed by the Lamotte Company.⁶ A sample of 10 mL water was taken from the potable container where households stored their drinking water. A sterile water sampling bag with a dechlorinating agent (2–2197) was provided with the test kit to collect the sample and neutralize⁷ any chlorine which may be present in the water. After collection, the sample water was sealed and incubated in an oven at a temperature of 35-degree Celsius in the tubes provided with the test kit package. A UV light was then used to test the presence of the *E. coli* bacteria in the water after a period of 48 h. The presence of *E. coli* is confirmed if the water sample fluoresced with exposure to the UV light. Fig. 1 shows the testing procedure of the *E. coli* bacteria in the water sample.

3.1. Variable description

3.1.1. Outcome variables

This study uses two outcome variables. The first dependent variable, *self-reported health status (Self)*, is a categorical variable measured on three levels. The individuals were asked to rate their health status, and the response was measured as *Poor* = 1, *Medium* = 2, and *Good* = 3. The other dependent variable, *actual health outcome*, is a count variable measured in terms of the number of household members affected by diarrhea during the last 30 days. The *actual health outcome* was recorded on three categories: *Child*, *Adult*, and *Both*. This research uses both indicators of health (i.e., *self-reported health status* and *actual health outcome*) to increase robustness in the measurement of health outcomes. Self-reported health status is a subjective concept and can vary between individuals, even among those that have a similar health condition (i.e., the issue of bias from multiple factors). We argue that the latter measure mentioned above, the count of household members affected by diarrheal incidences, has the potential to fit as a more accurate outcome variable compared to the self-reported health.

3.1.2. Behavior and infrastructural practices

The variables used to capture the behavioral and infrastructural practices include *Hand Washing*, *Utensils Washing*, *Deep Tubewell*, and *Flush Toilet*. *Deep Tubewell*⁸ is a binary variable that equals 1 if the household relies on a deep tubewell as their primary source of water, and 0 for all other sources of water. *Flush Toilet* is another binary variable that equals 1 if the household has a flush toilet system in their dwelling. If the respondent washes their hand each time after using the toilet then *Hand Washing* takes a value of 1; otherwise, it takes a value of 0. *Utensils Washing* refers to the utensil washing behavior of the households' water storage container. If the respondent washes their water storage container once a day, then it is coded as 1; otherwise, it takes a value of 0. Households usually tend to store water in their container on a one-day cycle and clean the containers before pouring water into it. The water sample to conduct the *E. coli* test was thus collected from the container where the households stored their water rather than from the point source itself. While we did not use the water sample from the point source for the *E. coli* test, previous studies have suggested that *E. coli* contamination is equally likely to occur from water storage containers [49]. The variable of interest in our study is *E. coli*, which equals 1 if the sampled water from a household tested positive for fecal contamination, and 0 if it did not test positive.

3.1.3. Averting behavior

We used the water boiling habits of the households to capture their averting behavior. *Boil* is a binary variable that equals 1 if the

⁴ VDC refers to Village Development Committee. This is the lowest administrative strata in Nepal.

⁵ The full survey consisted of questions to understand the environmental and health concerns as well as the water handling information of households in Nepal. A more detail information on the survey methods and other relevant information is presented in Kunwar & Bohara, [44]; and Kunwar et al. [47].

⁶ Available at: <http://www.lamotte.com/en/drinking-water/microbiological-testing/4-3616-uv.html>.

⁷ Water samples with chlorine residuals tend to suppress the growth of coliform bacteria when used with this kit. That is why a sterile water sampling bag with a dechlorinating agent (2–2197) is provided to collect the sample and neutralize any chlorine which may be present in the water.

⁸ Deep tubewell is considered as one of the safest sources of water in the study area. A deep hole is drilled into the ground (usually more than 150 m) with a hand pump attached to the top (UNHCR) [48]. This source of water is considered safe because of the depth from which the water is extracted. At a level of 150 m, it is very unlikely any contaminants can survive, and the water is also free from any human activity.

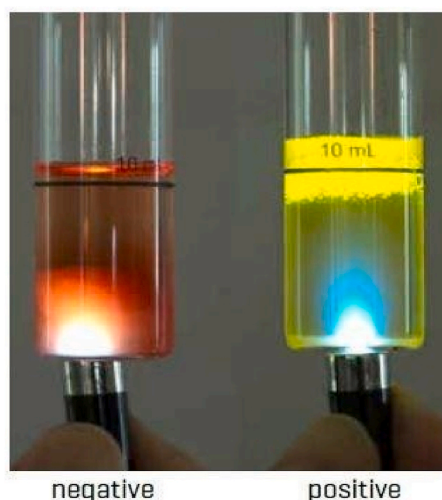


Fig. 1. *E. coli* bacteria testing procedure.

household boils their water before drinking, and 0 for everything else. This variable has been used to capture the averting mechanism practiced by the households to reduce the *E. Coli* risk in the drinking water.

3.1.4. Socioeconomic and demographic

The socio-demographic variables are captured by age, gender, education level, household income, and the water boiling behavior of the households. *Age* is a continuous variable and represents the age of the respondent, while *AgeSq* is the quadratic specification of the *Age* variable. *Male* is a binary variable which equals 1 if the respondent is a male. *Education* is a categorical variable that ranges from 1 to 4, where 4 is the highest level of education reached. The categories for *Education* are defined as: less than 10th grade = 1, equal to or above 10th grade and completed 12th grade = 2, completion of bachelor's degree = 3, and completion of master's degree = 4. The *Income* variable captures the total household income and is derived using an asset index following the methodology by Kunwar & Bohara [44]. This index comprises of 5 quintiles with the highest quintile representing the richest households and lowest quintile for the poorest households.

Table 1 presents the descriptive statistics of the households that responded to the survey. About 55-percent of the respondents reported their health status as “Medium” while only 18% described themselves as being in a “Good” status. The mean level of self-reported health for the entire sample is 1.90, which suggests that most of the individuals considered themselves to be relatively healthy. The number of children affected by diarrhea is captured by the *Child* variable, and its average value equals 0.17. The mean for the number of adults affected by diarrhea, *Adult*, is also 0.17, although the standard deviation is much higher compared to *Child*. The data indicates that 35% of the households' drinking water tested positive for the *E. coli* bacteria. Nicholson et al. [50] also found that 24% of households in Nepal were tested positive for *E. coli* contamination on their drinking water (see Table 2).

We used four variables to characterize the cleanliness behavior and hygiene infrastructure of households. On average, about 50% of the households used *Deep Tubewell* as a source of water in their homestead, and 64% of households use the *Flush Toilet* system.⁹ A large percentage of the respondents (71%) wash hands after using the toilet each time, and 61% of households wash their drinking water storage utensils every day. The average age of the respondents is 42.27. The majority of the respondents were male with the ratio of male to female percentage in the survey being 74%–26%.

In terms of the education level, more than half of the respondents (54.14%) have not completed their 10th grade education in high school. About 7% of the respondents have completed their bachelor's degree, while only 2.87% have obtained a master's degree. The distribution of education level of the sample in this study is relatively similar to other developing nations [51]. The mean level of *Income* quintile is 3.19, which indicates a moderate-income level of households on average. The left panel in Fig. 2 displays the relationship between average self-reported health status and the *E. coli* contamination in the drinking water. The figure indicates a negative relationship, on average, between self-reported health status and bacterial contamination in the drinking water. This indicates a better (worse) health status is associated with an absence (presence) of *E. coli* contamination in drinking water.

When we consider the number of household members affected by diarrhea, a similar pattern is evident from the right panel of Fig. 2. A higher percentage of diarrheal incidence is associated with *E. coli* contamination in the drinking water. The chi-square test

⁹ The percentage of households using *Flush Toilet* might seem a little high in the context of developing countries. However, our survey includes data from one municipality and 2 VDC's. The municipality experiences a better standard of living due to higher household income and the overall livelihood status compared to the two VDCs. Due to the proportional weight sampling while undertaking the survey, a large portion of the observation were drawn from the municipality. Hence, we see a high number of households employing the *Flush Toilet* system for their sanitation facility.

Table 1

Variable definition and descriptive statistics of the sampled households.

Variables	Definition	Mean	S.D.	Min	Max
Outcome Variables					
Self	Poor = 1 (27.4%) medium = 2 (54.5%) Good = 3 (18.1%)	1.9	0.66	1	3
Child	Number of children affected by diarrhea in the household in the last 30 days. The study considers children to be those that are less than 18 years of age.	0.17	0.48	0	2
Adult	Number of adults affected by diarrhea in the household in the last 30 days	0.17	0.61	0	8
Both	Number of total household members affected by diarrhea in the last 30 days. This is the summation of the child and adult variable.	0.34	0.82	0	10
<i>E. coli</i>	= 1 if <i>E. coli</i> is present in the drinking water, otherwise 0	0.35	0.47	0	1
Behavioral and Infrastructural					
Deep Tubewell	= 1 if household has deep tubewell; 0 for everything else	0.5	0.5	0	1
Flush Toilet	= 1 if household has flush toilet; 0 for everything else	0.64	0.48	0	1
Hand Washing	= 1 if the respondent washed hand each after using toilet; 0 for everything else	0.71	0.65	0	1
Utensils Washing	= 1 if the drinking water storage container is washed every day; 0 for everything else	0.61	0.48	0	1
Averting Behavior					
Boil	= 1 if the household boils the water before drinking, 0 for everything else	0.21	0.41	0	1
Socioeconomic and Demographic					
Age	Age of the respondent	42.27	14.96	18	85
AgeSq	Squared term of the Age	2010.3	1375.2	324	7225
Male	= 1 if the respondent is a Male, 0 for Female	0.74	0.43	0	1
Education	Less than 10th grade = 1 (54.14%) 10th grade to 12th grade = 2 (35.99%) Bachelors Complete = 3 (7.01%) MS Complete = 4 (2.87%)	1.58	0.74	1	4
Income	Monthly income of the households.	3.19	1.37	1	5
Number of HH Member	Number of HH members in each household	7	4	1	28
Observations		314			

Table 2Probit estimates and marginal effects of cleanliness behavior and hygiene infrastructures on *E. coli* contamination in drinking water.

	(1)	(2)	(3)	(4)
			Marginal Effect	
VARIABLES	<i>E. coli</i>	<i>E. coli</i>	<i>E. coli</i>	<i>E. coli</i>
Hygiene Infrastructure				
Deep Tubewell	-0.422** (0.178)	-0.355** (0.174)	-0.110* (-2.42)	-0.0942* (-2.07)
Flush Toilet	-0.854*** (0.185)	-0.881*** (0.184)	-0.222*** (-5.07)	-0.234*** (-5.33)
Cleanliness Behavior				
Handwashing	-0.783*** (0.185)	-0.728*** (0.180)	-0.204*** (-4.64)	-0.193*** (-4.39)
Utensils Washing	-1.003*** (0.176)	-0.989*** (0.174)	-0.261*** (-6.70)	-0.263*** (-6.69)
Averting Behavior				
Boil	-0.179 (0.214)	-0.168 (0.212)	-0.0467 (-0.84)	-0.0446 (-0.79)
Socio-Demographic				
Age	0.0460 (0.0332)	0.0448 (0.0329)	0.0120 (1.40)	0.0119 (1.37)
AgeSq	-0.000487 (0.000361)	-0.000495 (0.000357)	-0.000127 (-1.36)	-0.000131 (-1.39)
Male	-0.0703 (0.199)	-0.0335 (0.195)	-0.0183 (-0.35)	-0.00890 (-0.17)
Education	0.270** (0.123)	0.236* (0.121)	0.0703* (2.23)	0.0625* (1.97)
Income	0.0875 (0.0713)	0.0811 (0.0692)	0.0228 (1.24)	0.0215 (1.18)
Constant	-0.213 (0.745)	-0.128 (0.738)		
Observations	314	314	314	314
Fixed Effect: Urban -rural	Yes	No	Yes	No
AIC	315.92	318.37	-	-

Note: Standard errors (robust) in parentheses for column 1–2 and *t* statistics in parentheses for column 3–4.****p* < 0.01, ***p* < 0.05, **p* < 0.1.

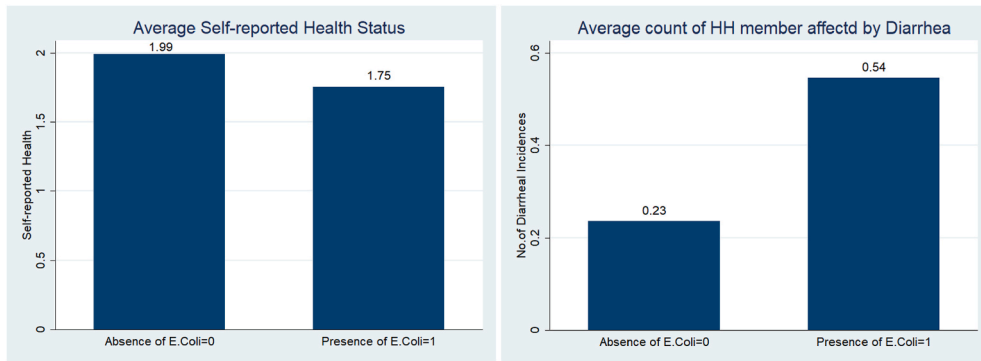


Fig. 2. Relationship between *E. coli* contamination and health indicators.

supports the result from Fig. 2 and suggests a significant dependence between self-reported health and *E. coli* with a p-value of 0.009. The chi-square test and the corresponding p-value also reveals a significant association when we consider diarrheal incidences in the household. Fig. 3 depicts the relationship between the level of *E. coli* in the drinking water and the share of households practicing cleanliness behavior and using hygiene infrastructures. It is apparent that households with no bacterial contamination in their water practice have a higher level of cleanliness behavior and hygiene infrastructure.

4. Empirical analysis

This section describes the empirical methods employed to estimate the effect of *E. coli* on health outcomes. The empirical specification below is motivated by the conceptual framework of the health production function approach first proposed by Grossman [52]. According to this approach, the health outcome, H_i is assumed to depend on the water quality (E), which itself depends on the cleanliness behavior (B) and infrastructure (I) of the household.

The health outcomes of individual i is assumed to be a function of the level of *E. coli*, the socioeconomic and demographic factors, and takes the following specification:

$$H_i = \beta_0 + X_i\beta + E_i\gamma + \eta_i + \varepsilon_i \quad (1)$$

where, H_i denotes the outcome variables which comprises of: (i) self-reported health status of the individuals; and (ii) the number of household members affected by diarrheal incidences in the past 30 days. E_i is a dichotomous variable that captures the outcome of the *E. coli* contamination test in the drinking water, and X is the vector of the socioeconomic and demographic factors, which includes the respondents' age, gender, education level, and the household income. β_0 , β , η_i and γ are the parameters to be estimated, and ε_i captures the error term. Since the survey area encompassed one municipality (urban) and two VDC's (rural), the parameter η_i is used as the urban-rural fixed effect to account for the variability in location¹⁰. The parameter β_0 is the intercept of the model, β is a vector of coefficients which measures the change in H_i due to a change in the socioeconomic and demographic factors, and γ stands for the impact on health outcomes from a change in the *E. coli* level.

The focus of this study is on measuring the causal effect of the level of *E. coli* bacteria in drinking water on health outcomes. As has been highlighted earlier, *E. coli* contamination in drinking water can occur from multiple factors,¹¹ two of them being cleanliness behavior and hygiene infrastructures. In fact, past studies have shown a direct relation between *E. coli* and cleanliness behavior [54–56]. These findings, however, neither confirm nor invalidate the argument that cleanliness behavior and hygiene infrastructures can also affect health outcomes. This idea sets the premise of our research. We argue that the impact of cleanliness behavior and hygiene infrastructure on health outcomes can be established via an intermediary process of bacterial contamination in the drinking water.

We use the *E. coli* level in drinking water as a mediator to link the interdependence between cleanliness behavior, hygiene infrastructure, and health outcomes. However, the exogenous use of *E. coli* in Equation (1) can potentially deliver inconsistent estimates. Furthermore, it is also possible that unobservables like the public drinking water source might influence the outcome of *E. coli* level among households that depend on it relative to those that use other water sources [57].¹² These arguments collectively establish

¹⁰ The settlement characteristics of the urban area is much different than rural areas. We expect that the coefficients will have variability based on the difference of location. We add an urban-rural fixed effect to control for this variability.

¹¹ Other factors that can affect the *E. coli* contamination in drinking water are mode of transportation from source point to storage [3], animal ownership [40], natural factor such as precipitation [53] etc.

¹² In our sample, 16% of the households obtain water for the public tap. As suggested from the previous literature, this can also affect the formation of *E. coli*, which does not fall in the cleanliness behavior of hygiene infrastructure practiced by the households. In other words, *E. coli* may increase as a result of water quality in the water drinking system, not households' behavior. Consequently, this can be a source of omitted variable bias and resulting endogeneity.

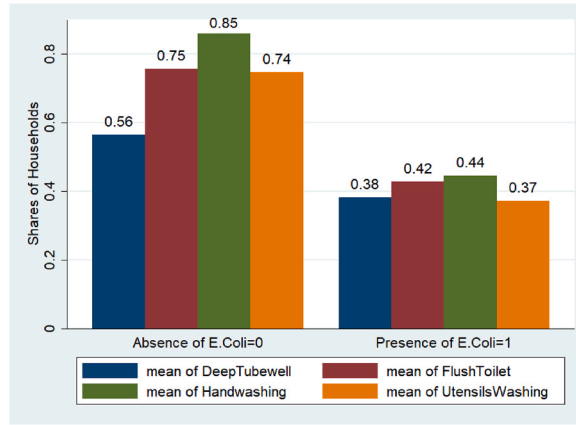


Fig. 3. Relationship between *E. coli* contamination, cleanliness behavior, and hygiene infrastructure.

that the *E. coli* variable might suffer from an endogeneity issue. Failing to account for endogeneity can result in a biased outcome as it violates the first Ordinary Least Squares (OLS) assumption, $COV(E_i, \varepsilon_i) = 0$ [58]. To incorporate the potential for endogeneity issues in our data, Equation (1) is reconstructed by a two-stage reduced form system and estimated by a Control Function (CF) approach. The first-stage can be stated as:

$$E_i = \alpha_0 + X_i\beta + Z_i\delta + \eta_i + V_i \quad (2)$$

where, α_0 is the constant, Z is a vector comprising of variables to capture the behavioral (*Utensils Washing*, *Hand Washing*) and infrastructural (*Deep Tubewell*, *Flush Toiler*) factors, and V_i is the error term. The variables comprising in vector Z are employed only in Equation (2); hence, it identifies the specification of the second stage regression. The dependent variable in the first-stage in Equation (2) is a dichotomous variable and is estimated by a probit model. However, the dependent variables of the second stage in Equation (1) are categorical (*self-reported health status*) and count (*number of household members affected by diarrheal diseases*) variables. Therefore, a two-stage least Square (2SLS) approach does not account for all these issues and is unable to produce consistent estimates [58]. Accordingly, a CF approach is employed to overcome this limitation. The relation between the two error terms is specified by a linear relationship to capture the possible endogeneity, as stated below:

$$\varepsilon_i = V_i\theta + \tau_i \quad (3)$$

where θ captures the marginal effect of V_i on ε_i . τ_i follows a normal distribution ($\tau_i \sim N(0, 1)$) with the condition $E(V_i, \tau_i) = 0$. Given the linear relationship between the error terms, and an assumption of independence between (X, Z) and V_i , the CF approach estimates the presence of *E. coli* by a probit model. The probit model can be specified as:

$$P(E_i = 1|X, Z) = \Phi(\alpha_0 + X_i\beta + Z_i\delta + \eta_i) = \Phi(\Omega K) \quad (4)$$

where, K is a vector containing the variables from X and Z . A generalized residual is calculated after estimating Equation (4) by the following formula:

$$g_i = [E_i * \lambda(\Omega K) - (1 - E_i * \lambda(-\Omega K))] \quad (5)$$

where, $\lambda(\Omega K)$ is the Inverse Mills Ratio (IMR), and is given by $\varphi(\Omega K)/\Phi(\Omega K)$. The IMR is included to control for the possible endogeneity of the *E. coli* variable [24]. Once the generalized residual function is calculated, Equation (1) can now be re-written as:

$$H_i = \beta_0 + X_i\beta + E_i\gamma + g_i\sigma + \eta_i + \mu_i \quad (6)$$

Here, g_i is the generalized residual, σ measures the magnitude of the possible endogeneity, and μ_i is the error term. The generalized residual, g_i , also serves as an explanatory¹³ variable, and we assume that $COV(E_i, \mu_i) = 0$. The main rationale for the inclusion of the generalized residual is to control for the possible endogeneity of *E. coli*. The identification requirement for the CF approach is nonzero¹⁴ coefficients of Z in the first-stage regression (i.e., $\delta \neq 0$ in Equation (2)). Ordered probit regression is used to estimate the self-reported health status, and a negative binomial model is employed to estimate the actual health outcome. While using the number

¹³ The rationale behind this residual plugging-in approach is to establish an uncorrelation with E_i and ε_i . Because ε_i and V_i are uncorrelated with K , τ_i is also uncorrelated with K . Consequently, τ_i is also uncorrelated with E_i . If this relationship holds, ε_i can't influence E_i . Therefore, we can get valid and consistent estimates from Equation (6), given the assumption $COV(E_i, \mu_i) = 0$ holds.

¹⁴ A simple F-test of joint significance is used to test this requirement and presented in the result section. Furthermore, the estimated value of σ produces a heteroskedasticity-robust Hausman test statistic, which determines the possible endogeneity of *E. coli*.

of affected members by the diarrheal disease across households, we control for the number of household members as an exposure variable in the negative binomial model [58,59]. The negative binomial model takes the following specification:

$$\mu_i = \exp[\ln(\text{Number of HH Member}) + \beta_0 + X_i\beta + E_i\gamma + g_i\sigma + \eta_i] \quad (7)$$

where, μ_i is the mean of H_i . All regression models presented in this study are estimated with and without urban-rural level fixed effects. In addition to the two-stage estimation, an OLS estimation is also reported as an additional robustness check.

5. Results and discussion

This study focuses on the effect of *E. coli* contamination in drinking water on the health outcomes in the context of a developing country. Our findings are in line with the existing research on health status and bacterial contamination, and we further add upon the past literature by highlighting the role of intermediary processes in explaining health outcomes. Table (2) presents the probit estimation and the marginal effects for the effect of cleanliness behavior, hygiene infrastructure, and socioeconomic factors on *E. coli* contamination. These estimates correspond to the result of Equation (2) in the empirical analysis section. Columns (1) and (2) present the result of the probit estimation, while columns (3) and (4) show their respective marginal effects. We account for the urban-rural fixed effect in column (1), while column (2) excludes the urban-rural fixed effect. The results from both columns in Table (2) confirm the finding in figure (3): an increase in the practice of cleanliness behavior and hygiene infrastructure decreases the likelihood of *E. coli* contamination in the drinking water.¹⁵

The variables capturing the cleanliness behaviors and hygiene infrastructures are significant and suggest a negative effect of bacterial contamination in drinking water. Moreover, these effects are consistent across both the urban and rural areas, as evident from the fixed effects in Table (2). *Utensils Washing* has the largest marginal effect among the cleanliness and infrastructural factors. The probability of bacterial formation in the drinking water decreases by 26-percentage points if households wash their drinking water storage utensils every day. On the other hand, the source of the drinking water, *Deep Tubewell*, has the least impact on bacterial contamination. The probability of *E. coli* contamination decreases by 11-percentage points if households use water from deep tubewell for drinking purpose. A similar result was found by Ostwald et al. [60] in understanding the factors behind fecal contamination in drinking water in Lima, Peru. Their results identified that post-source contamination of drinking water was largely influenced by the hygiene behaviors of the households. Brick et al. [42] also found a similar result where they identify that contamination chances in drinking water increases during the storage periods.

In Table (2), we also find that the boiling behavior of the household is negatively¹⁶ associated with *E. coli* contamination; however, the variable itself is not significant. The boiling behavior of the household has not been seen as a significant factor affecting *E. coli* level in other studies too. Ostwald et al. [60] find no significant association between boiling behavior of the households in Lima, Peru, and the level of *E. coli* contamination in their drinking water. The socioeconomic variables¹⁷, for the most part, are not significant and do not suggest any potential effect on the formation of *E. coli* bacteria in drinking water. The findings of the impact of *E. coli* on health outcomes and their associated marginal effects are presented in Table (3) – Table (7). These estimates correspond to the result of Equation (6) in the empirical analysis section.

Table (3) presents the ordered probit estimates of the health outcome when we consider the self-reported health status. The respondents' self-reported health is captured by the categorical variable *self*. Column (1) in Table (3) uses only the *E. coli* variable and does not control for any other factors or the fixed effect, whereas columns (2) and (3) present the full model with the full controls.¹⁸ The latter two columns (2 and 3) also use a generalized residual calculated from the first stage. The results indicate a negative and significant relationship of *E. coli* contamination with self-reported health. An increase in the likelihood of *E. coli* contamination in drinking water reduces the likelihood that respondents report their health as "Good".

Overall, our result suggests that the variables capturing the cleanliness behaviors and hygiene infrastructures are significant and suggest a negative effect of bacterial contamination, a finding that has been established in other studies too. For instance, Kile et al. [29] investigate the linkages between drinking water quality and self-reported maternal health in Bangladesh. Unlike our study, which uses *E. coli* contamination in the drinking water, their study used arsenic exposure to drinking water as an indicator of water quality, and their findings indicated a significant deterioration of self-reported maternal health status from exposure to arsenic contamination.

¹⁵ We also performed a multicollinearity test by estimating the Variance Inflating Factor (VIF) for the explanatory variables of this regression. The VIF test did not indicate any significant dependence among explanatory variables. The VIF's for each variable were at the accepted standard (VIF < 10).

¹⁶ In a related study to ours, Kunwar and Bohara [44] find that households with absence of *E. coli* in their drinking water were more likely to boil their water. The direction between *E. coli* and boiling behavior in our study and the study by Kunwar and Bohara [44] is similar, although we do not find the relationship to be significant. This might be because we seek to understand the factors that affect *E. coli* contamination, whereas the other study looks at factors that affect water averting behaviors. Thus, it is possible that households that do not have *E. coli* are those that are more likely to boil their water, whereas households that boil their water may or may not have *E. coli* contamination.

¹⁷ Though education is significant in first stage regression, it is not significant in any specifications on the main regression. This result is unexpected and not easily explainable given the scope of this study. Thus, the significance of the education variable is not discussed.

¹⁸ In Table 3, we wanted to show that E.Coli has a significant effect on health status even if we don't control for other variables. Consequently, we estimate with all the variables in the second column and still find the significance of E.Coli. In this way, we can conclude the robustness of the result found.

Among the socio-demographic factors in Table (3), household income is the only significant variable and suggests that households with higher income are likely to rate their health as better. While the findings from Tables (2) and (3) indicate clear linkages between hygienic behavior, bacterial contamination, and health outcome, the self-reported health ranking used to capture the health outcome so far is a subjective measure and can lead to biased results due to heterogeneity among the individuals.¹⁹

Column (1) presents the result of the probit model of *E. coli* formation in drinking water with an Urban-rural fixed effect. The dependent variable is the dichotomous *E. coli* variable. Column (2) presents the results without a fixed effect. Column (3) presents the marginal effect of the probit results of column (1). Column (4) presents the marginal effect of the probit results of column (2). The marginal effect is calculated for continuous variable by $\frac{\partial \Pr(E.Coli = 1)}{\partial x_k} = \varphi(x\beta) \beta_k$ and $\frac{\partial \Pr(E.Coli = 1)}{\partial x_k} = \Phi(x_1\beta) - \Phi(x_0\beta)$ for binary variable. Here, φ is the standard normal p. d.f and Φ is the c. d.f [58].

Column (1) presents the ordered probit estimates of Self-reported health status with only *E. coli* as the independent variable. *Self* is the dependent variable. Column (2) presents the estimates for *Self* with the full control variables. gfe^1 is the generalized residual calculated from the column (1) of table (2) which includes a fixed effect. Column (3) presents the result without a fixed effect. gr^2 is the generalized residual calculated from column (2) of table (2). Both gfe and gr follow the formula of equation (5) in the empirical section.

To account for the potential biases in the measurement of health outcome, a negative binomial model is also estimated where we employ the prevalence of diarrhea as an indicator of health in relation to the quality of drinking water. Table (4) shows the negative binomial estimation of the number of household members affected by diarrheal incidence in the past 30 days across three categories: *Child*, *Adult*, and *Both*. These coefficients in Table (4) also capture the marginal effects. We use the number of household members as an exposure variable in each of the three models. All specifications in Table (4) indicate a strong positive association between the number of people affected by diarrhea in each category and *E. coli* contamination in their drinking water. The likelihood of *E. coli* in the drinking water increases the expected incidences significantly across all specifications.

Columns (1), (3), and (5) in Table (4) show the negative binomial estimate of the *child*, *adult*, and *both* respectively by controlling for the urban-rural fixed effect, while columns (2), (4), and (6) do not account for the urban-rural fixed effect. Due to the unavailability of the child's age and gender, the variables to capture the age and gender of the child (i.e., *Age*, *AgeSq*, and *Male*) are excluded from the analysis. The result in column (1) suggests that the difference in the expected incidences is about 0.932 units higher when *E. coli* is present in the water among the *children* group. The difference in expected incidences is expected to be 0.967 units higher when we consider the *adults*, whereas the difference in the expected incidences is expected to be 0.913 units higher when the outcomes for adults and children are combined. A similar finding has been established in other studies as well. Amr & Yassin [61] found a strong association between *E. coli* in the water system and diarrheal incidence in both historical and self-reported data among households in the Gaza Strip, Palestine. Other studies that have found parallel results to ours include those by Ercumen et al. [14] and Fagerli et al. [15].

Among other factors associated with *E. coli* contamination, the variable *Boil* has the expected sign in every specification, but it is significant only for *adults*. The variables that capture the age, sex, and education level are not significant in determining the diarrheal prevalence, while the income variable suggests a negative relationship across the *Adult* and *Both* categories. The parameter ' gfe ' in Table (4) is the generalized residual calculated from column (1) in Table (2), and it includes the fixed effect. The parameter ' gr ' is the generalized residual calculated from column (2) in Table (2). ' gno ' is the generalized residual, which is calculated by excluding the *Age*, *AgeSq*, and *Male* variables, and by incorporating the fixed effect. Lastly, the parameter ' $gnofe$ ' is the generalized residual calculated without controlling for *Age*, *AgeSq*, and *Male* and without incorporating the fixed effect.

These generalized residual coefficients provide the heteroskedasticity-robust Hausman test for endogeneity. A failure to reject the null hypothesis of the endogeneity of *E. coli* indicates the consistency of our estimates. Our results remain consistent even when excluding the fixed effect in column (2), (4), and (6), which validates the robustness of the findings. Each column in Table (4) also reports a chi-square value, which is the output of the likelihood ratio test for comparison between the Poisson and the negative binomial process. We reject the null hypothesis of Poisson regression based on the P-value of the likelihood ratio test in each column, thereby validating the negative binomial model as the proper estimation strategy for our data.²⁰

Columns (1), (3), and (5) show the negative binomial estimate of the *child*, *adult*, and *both* respectively using the urban-rural fixed effect. Columns (2), (4), and (6) show the negative binomial estimate of the *child*, *adult*, and *both*, respectively, without the urban-rural fixed effect. *Child*, *Adult*, and *Both* are the dependent variables. gfe^1 is the generalized residual calculated from column (1) of table (2), which includes a fixed effect. gr^4 is the generalized residual calculated from column (2) of table (2). gno^2 is the generalized residual calculated without controlling for *Age*, *AgeSq*, and *Male* of the child but including the fixed effect. $gnofe^3$ is the generalized residual calculated without controlling for *Age*, *AgeSq*, and *Male* of the child and without the fixed effect. All the generalized residuals are calculated from equation (5) of the empirical section. The coefficients are on a natural logarithm scale. These coefficients are also the marginal effects.

Table (5) shows the output for the marginal effects of the ordered probit and the negative binomial estimates. Columns (1–3) present the estimates for the self-reported health status, *Self*, for the three categories: *Poor*, *Medium*, and *Good*; while columns (4–6) present the marginal effects of the negative binomial model for the *Child*, *Adult* and *Both* samples. The result in Table (5) provides clear

¹⁹ Since boiling is not the main concern in this study, we did not include it under cleanliness behavior or hygienic infrastructure.

²⁰ We have also estimated the equation with an urban dummy variable (*Location* = 1 if it is Urban, *Location* = 0 if it is Rural) instead of using the fixed effects. This approach also does not change the result in terms of the coefficient value or significance level. For this reason, we decided to keep the fixed effect in all the tables after Table 4. We wanted to check whether the result is sensitive to the urban vs. rural distinction. As it is not, we proceeded with only a fixed effect.

Table 3

Ordered Probit estimates of self-reported health status.

	(1)	(2)	(3)
VARIABLES	<i>Self</i>	<i>Self</i>	<i>Self</i>
<i>Ecoli</i>	−0.402*** (0.133)	−0.416*** (0.147)	−0.422*** (0.143)
<i>Boil</i>		−0.221 (0.152)	−0.222 (0.154)
<i>Age</i>		−0.0292 (0.0288)	−0.0286 (0.0280)
<i>AgeSq</i>		0.000333 (0.000311)	0.000336 (0.000303)
<i>Male</i>		0.170 (0.161)	0.147 (0.157)
<i>Education</i>		0.00524 (0.105)	0.0161 (0.102)
<i>Income</i>		0.185*** (0.0471)	0.184*** (0.0454)
<i>gfe</i> ¹		0.0400 (0.162)	
<i>gr</i> ²			0.00229 (0.161)
<i>Constant 1</i>	−0.752*** (0.0904)	−0.660 (0.677)	−0.643 (0.648)
<i>Constant 2</i>	0.785*** (0.0895)	0.945 (0.686)	0.955 (0.656)
Observations	314	314	314
Fixed Effect: Urban-rural	No	Yes	No

Note: Standard errors (robust) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

evidence of negative association between *E. coli* and self-reported health as the health ranking of the respondent increases. The change in the predicted probability of reporting the health status as “Good” decreases significantly by 10-percentage points. The effect on the “Medium” category decreases by two²¹ percentage points, although the variable itself is not significant. In the lowest category, the effect on the “Poor” category increases by 13-percentage points, which is positive and significant.

The survey area in this study includes an urban municipality and two rural VDC's, with a substantial portion of the data drawn from the urban area (i.e., Siddharthanagar Municipality). Of the 314 observations, 266 observations (84%) are drawn from the municipality, and 48 observations from the Bashantapur and Bagha VDC's. In addition to a large sample of the respondents belonging to the urban area, the amount of bacterial contamination in drinking water is also high in the urban area. Out of 266 observations, 95 households (35%) drinking water tested positive for *E. coli* contamination, while only 31% tested positive in the rural areas. As a robustness check, we estimate the effect of *E. coli* on only the urban sample to observe the consistency of our results. We follow the same estimation methodology in the urban sample that was used for the entire study area. Since the marginal effects of the ordered probit estimates show the precise effect, we present the marginal effects in each category. For estimation consistency, the *Number of HH Member* is still used as the exposure variable in all the negative binomial models presented in Table (6).

Column (1–3) shows the marginal effect of *Self* in three different categories such as poor = 1, medium = 2, and good = 3. Column (4–6) shows the marginal effect of the negative binomial models. Fixed effect is considered in all the models. *gfe*¹ is the generalized residual calculated from the column (1) of table (2) which includes a fixed effect. *gno*² is the generalized residual calculated without controlling for *Age*, *AgeSq*, and *Male* of the child but including the fixed effect. All the generalized residuals are calculated from equation (5) of empirical section.

Column (1–3) shows the marginal effect of *Self* in three different categories such as poor = 1, medium = 2, and good = 3 based on the samples collected from the Siddharthanagar Municipality. Column (4–6) shows the marginal effect of the negative binomial models. *gr*¹ is the generalized residual calculated from column (2) of Table (2). *gnofe*² is the generalized residual calculated without controlling for *Age*, *AgeSq*, and *Male* of the child and without the fixed effect.

The general findings in Table (6) are similar to the results in Table (5) and supports the consistency of our results even when the Siddharthanagar sample is considered by itself. The change in the predicted probability on each category for self-reported health is consistent between Table (5) and (6). However, the difference in the expected incidence in the Siddharthanagar sample is dissimilar when compared to the results of the entire sample. For instance, the impact on the number of children affected by diarrhea in Siddharthanagar is lower than the entire sample, while the impact on the number of adults shows a larger estimate. The Chi-Square statistic and the corresponding P-value indicates the fit of the data by a negative binomial process. As mentioned earlier, several

²¹ The effect in the medium category is small because it indicates the net effect of *E. coli* increase. The 2-percent change addresses the net change in the predicted probability of moving to “Medium” from the “Good” category, and then to the “Poor” category as *E. coli* increases.

past studies have indicated a direct association between *E. coli* contamination and health outcomes, although these studies do not account for any intermediary process [11–15].

As another robustness check for the validity of the results, and also to confirm the findings of previous research, we estimate the direct association between *E. coli* contamination and health outcomes without controlling for the endogeneity of the *E. coli* variable. This is done by directly estimating Equation (1) by employing the ordered probit and the negative binomial model, and the output is shown in Table (7). The general results are still consistent and support the main findings of the paper: the likelihood of *E. coli* contamination in drinking water reduces the self-reported health status and increases the expected incidence of diarrhea among the household members. Similar to the previous findings, only the *Boil* variable is significant and only in the *Adult* regression, and *Income* is significant with the expected sign.

Column (1) presents the ordered probit estimates of Self-reported health status. *Self* is the dependent variable. Column (2–4) presents the estimates for *child*, *adult*, and *both* by negative binomial models. *Constant 1* and *Constant 2* are the two constants of the ordered probit model where *Constant* is the constant for the negative binomial model.

We have used both self-reported health and the number of household members affected by diarrheal incidences as indicators of the health status. The results show consistency between the two indicators of health status. As indicated earlier, self-reported health status suffers from the limitation of bias while reporting it [62,63]. However, studies [64,65] have also shown concordance between generic self-reported health and a specific measure. The results from negative binomial regressions from this study also suggest that self-reported health could be used as an indicator of health due to the consistency of results. Since the measurement between the two health indicators is different in this study, we do not discuss whether the estimates of *E. coli* from the ordered probit and negative binomial regressions can be used as substitutes.

The negative binomial estimates show a higher value of the difference between the expected incidences for adults than child (Tables 4 and 7). This suggests that *E. coli* contamination has a higher impact on the health of an adult than a child. Research has shown that intestinal parasites (such as *Giardia*) are dominant in causing diarrhea than bacterial contamination, such as *E. coli* among children. Sherchand et al. [66] investigated the stool sample of 440 diarrheal child patients in Nepal and found that intestinal parasites (36.5%) and pathogenic bacteria (20%) were the cause of most of the diarrheal infections. Among the pathogenic bacteria, the predominant bacteria were *Shigella species* (36.8%), *Vibrio species* (26.3%), *Escherichia coli* (22.8%), and *Salmonella species* (14.03%) respectively. Their finding further supports our results of low impact on a child from *E. coli* contamination since other parasites may play a dominant role in the prevalence of diarrhea compared to adults. Moreover, as a measure of another robustness check, we used the number of child and the number of adults as exposure variables and re-estimated the regressions of *Child* and *Adult* in each of the specifications in Tables 4–7. We find the consistency of the results in terms of the sign of coefficients and level of significance; however, we do not report the results here to limit the length of the paper.

Finally, the results from this study also indicate that boiling the water alone may not protect from *E. coli* contamination, whereas washing utensils could have a significant impact on reducing bacterial contamination in drinking water. Among other covariates, the marginal effect of income (*Income*) is significant with the expected sign on categories “*Poor*” and “*Good*” (Table 5). The predicted probability for self-reported health status, *Self*, decreases by five percentage points if the household is considered “*Poor*” while it increases by four percentage points if the household income is considered “*Good*”.

6. Conclusions

This study investigated the linkages between water quality level and two measures of health outcomes: self-reported health status, and the number of household members affected by diarrheal diseases. The novel contribution of this paper is that we introduce an intermediary process to understand the relationship between water quality level and health status. The study is carried out in three locations in the Rupandehi district of Nepal. The results not only confirm prior findings on the significant linkage between health outcomes and drinking water contamination but also extend the understanding of factors pertaining to *E. coli* formation in the drinking water. More specifically, our analyses from the primary survey data collected suggest that a) better hygiene infrastructure and cleanliness behavior reduces the likelihood of *E. coli* contamination, b) higher likelihood of *E. coli* presence significantly decreases the self-reported health status, c) higher likelihood of *E. coli* presence also significantly increases the number of diarrheal incidences. Furthermore, the outcome variables are estimated by accounting for the endogeneity of *E. coli*.

The results remain consistent across the various specifications carried out to test the robustness of our findings. While the general results from this paper suggest a link between health outcomes, water quality, and cleanliness and hygienic behaviors, these findings should still be taken with caution. For one, although we accounted for the potential endogeneity of the *E. coli* variable, the results might still be sensitive to unobservables despite the several validities and robustness checks conducted. Incorporating other variables in the estimation of *E. coli* was not possible since data was not collected on other possible sources of *E. coli* contamination. Similarly, information on the child’s age and gender could also have explained more of the actual outcome, but it was outside the scope of the study due to the limitation²² of the data.

The findings from this study can nevertheless provide substantial guidance to the Siddharthangar municipality for appropriate policy formulation to alleviate health hazards from water-related issues. One policy measure that could be implemented is to begin

²² While we do not think the central result of the study would differ, incorporating the children’s gender and age would probably make the findings more robust and comparison of the result between the two sub-groups (adults and children) would be more comparable since both groups would have similar control variables to capture the variations.

Table 4Negative binomial estimates of the number of household members affected by diarrhea among the *Child*, *Adult*, and *Both* samples.

VARIABLES	(1) <i>Child</i>	(2) <i>Child</i>	(3) <i>Adult</i>	(4) <i>Adult</i>	(5) <i>Both</i>	(6) <i>Both</i>
<i>Ecoli</i>	0.932*** (0.356)	0.914** (0.361)	0.967*** (0.349)	1.048*** (0.356)	0.913*** (0.247)	0.902*** (0.247)
<i>Boil</i>	−0.163 (0.432)	−0.176 (0.433)	−0.809 (0.527)	−0.886* (0.534)	−0.467 (0.330)	−0.477 (0.330)
<i>Age</i>			0.0452 (0.0674)	0.0493 (0.0690)	0.0477 (0.0485)	0.0475 (0.0484)
<i>AgeSq</i>			−0.000401 (0.000736)	−0.000431 (0.00075)	−0.000530 (0.000541)	−0.000536 (0.000541)
<i>Male</i>			−0.542 (0.372)	−0.577 (0.378)	−0.287 (0.269)	−0.285 (0.268)
<i>Education</i>	−0.109 (0.245)	−0.124 (0.249)	−0.277 (0.280)	−0.262 (0.281)	−0.198 (0.187)	−0.203 (0.186)
<i>Income</i>	−0.0789 (0.128)	−0.108 (0.129)	−0.249* (0.135)	−0.208 (0.136)	−0.157* (0.0918)	−0.145 (0.0916)
<i>gfe</i> ¹			0.628 (0.394)		0.285 (0.292)	
<i>gno</i> ²	−0.167 (0.439)					
<i>gnofe</i> ³		0.109 (0.417)				
<i>gr</i> ⁴				0.594 (0.408)		0.374 (0.285)
<i>Constant</i>	−3.722*** (0.590)	−3.606*** (0.581)	−3.840** (1.544)	−4.220*** (1.568)	−3.460*** (1.086)	−3.527*** (1.073)
Observations	314	314	314	314	314	314
Fixed Effect	Yes	No	Yes	No	Yes	No
Chi Square	7.82	10.45	12.89	17.69	18.00	18.19
P value	.003	.001	0.000	0.000	0.000	0.000

Note: Standard errors (robust) in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

regular monitoring of the water quality level of the water sources, which would make the municipal authorities more informed and better prepared to take protective actions to make the drinking water safe for households. In addition to municipalities monitoring the water sources, an alternative approach could be to train the households themselves to measure their water quality so they too can make informed decisions. This could be done by implementing programs such as “citizen science”²³ initiatives that allows the general people to become more aware and engaged to learn further about water cleansing behaviors and the risks of contaminants in their water. The results from this study could be instrumental in furthering research on water quality and health status in the developing world. For instance, based on our findings a randomized control trial (RCT) study could be carried out to analyze impacts on health status from changes in behavior and infrastructure among households with varying levels of information dissemination about their drinking water quality levels.

Our study finds that cleanliness behavior and hygiene infrastructures are the mitigating factors for reducing the outbreak of *E. coli* contamination in drinking water. Hence, another policy could be targeted towards nudging cleanliness habits while handling water, which could reduce health hazards from diarrheal incidences. Investment in raising awareness through social media campaigns and developing school-level curriculums on cleanliness practices, hygiene infrastructure, and water quality could all also aid in developing health knowledge among the general population. Finally, the construction of hygienic infrastructures, such as *deep tubewell* and *flush toilet*, incurs a significant amount of cost to the households. Subsidizing the costs of these infrastructures could induce a larger number of households to use them, which would thereby safeguard the entire community from health hazards like diarrheal incidences.

CRediT authorship contribution statement

Mohammad Mashiur Rahman: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Visualization. **Samrat B. Kunwar:** Funding acquisition, Writing – review & editing, Data curation, Resources. **Alok K. Bohara:** Project administration, Supervision, Conceptualization, Methodology, Validation.

²³ In fact, the Nepal Study Center at the University of New Mexico had recently implemented a “citizen science” program where the local volunteers in Siddharthanagar were trained to measure the contaminants in the nearby Danda River. This type of program could be extended to the household level as well. More information on this project can be found on Kunwar et al. [47], and also on: <https://foxc01.wixsite.com/yogdan/projects>.

Table 5
Marginal effects of *E. coli* on self-reported health and the diarrheal incidences.

	(1)	(2)	(3)	(4)	(5)	(6)
		Self				
	Poor = 1	Medium = 2	Good = 3	Child	Adult	Both
<i>Ecoli</i>	0.130** (2.90)	−0.0263 (−1.84)	−0.104** (−2.87)	0.932*** (0.356)	0.967*** (0.349)	0.913*** (0.247)
<i>Boil</i>	0.0691 (1.47)	−0.0140 (−1.37)	−0.0552 (−1.43)	−0.163 (0.432)	−0.809 (0.527)	−0.467 (0.330)
<i>Age</i>	0.00910 (1.01)	−0.00184 (−0.96)	−0.00726 (−1.00)		0.0452 (0.0674)	0.0477 (0.0485)
<i>AgeSq</i>	−0.000104 (−1.07)	0.0000211 (0.99)	0.0000831 (1.07)		−0.000401 (0.000736)	−0.000530 (0.000541)
<i>Male</i>	−0.0530 (−1.05)	0.0107 (0.93)	0.0423 (1.06)		−0.542 (0.372)	−0.287 (0.269)
<i>Education</i>	−0.00164 (−0.05)	0.000331 (0.05)	0.00130 (0.05)	−0.109 (0.245)	−0.277 (0.280)	−0.198 (0.187)
<i>Income</i>	−0.0576*** (−3.86)	0.0117 (1.95)	0.0460*** (3.96)	−0.0789 (0.128)	−0.249* (0.135)	−0.157* (0.0918)
<i>gfe</i> ¹	−0.0125 (−0.25)	0.00253 (0.24)	0.00996 (0.25)		0.628 (0.394)	0.285 (0.292)
<i>gno</i> ²				−0.167 (0.439)		
<i>N</i>	314	314	314	314	314	314
<i>Fixed Effect</i>	Yes	Yes	Yes	Yes	Yes	Yes

Note: *t* statistics in parentheses for column 1–3 and standard error in parentheses for column 4–6.

p* < 0.05, *p* < 0.01, ****p* < 0.001.

Table 6
Marginal effects of *E. coli* on Self-reported health and the diarrhea incidences for the Siddharthanagar municipality.

	(1)	(2)	(3)	(4)	(5)	(6)
		Self				
	Poor = 1	Medium = 2	Good = 3	Child	Adult	Both
<i>Ecoli</i>	0.123** (2.45)	−0.0216 (−1.44)	−0.102** (−2.42)	0.722* (1.69)	1.075*** (2.98)	0.880*** (3.23)
<i>Boil</i>	0.0349 (0.64)	−0.00612 (−0.58)	−0.0288 (−0.65)	−0.0233 (−0.05)	−0.678 (−1.28)	−0.345 (−0.97)
<i>Age</i>	0.00614 (0.70)	−0.00108 (−0.67)	−0.00506 (−0.69)		0.0461 (0.67)	0.0495 (0.95)
<i>AgeSq</i>	−0.0000668 (−0.69)	0.0000117 (0.65)	0.0000551 (0.69)		−0.000360 (−0.48)	−0.000475 (−0.83)
<i>Male</i>	−0.0633 (−1.05)	0.0111 (0.85)	0.0522 (1.06)		−0.354 (−0.92)	−0.197 (−0.67)
<i>Education</i>	−0.00854 (−0.22)	0.00150 (0.22)	0.00704 (0.22)	−0.211 (−0.67)	−0.212 (−0.76)	−0.232 (−1.11)
<i>Income</i>	−0.0570*** (−3.40)	0.00999 (1.65)	0.0470** (3.20)	−0.117 (−0.76)	−0.293* (−2.11)	−0.212* (−2.09)
<i>gr</i> ¹	0.0232 (0.48)	−0.00407 (−0.48)	−0.0191 (−0.48)		0.530 (1.31)	0.197 (0.62)
<i>gnofe</i> ²				−0.220 (−0.42)		
<i>N</i>	266	266	266	266	266	266
<i>Chi-square</i>				8.03	12.07	15.48
<i>P-Value</i>				0.002	0.000	0.000

Note: *t* statistics in parentheses for column 1–3 and standard error in parentheses for column 4–6.

p* < 0.05, *p* < 0.01, ****p* < 0.001.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 7Ordered probit and negative binomial estimates of *E. coli* without accounting for endogeneity.

VARIABLES	(1) <i>Self</i>	(2) <i>Child</i>	(3) <i>Adult</i>	(4) <i>Both</i>
<i>E. coli</i>	−0.421*** (0.140)	0.949*** (0.335)	1.158*** (0.345)	1.002*** (0.235)
<i>Boil</i>	−0.222 (0.153)	−0.184 (0.432)	−0.895* (0.529)	−0.500 (0.329)
<i>Age</i>	−0.0286 (0.0277)		0.0607 (0.0677)	0.0517 (0.0481)
<i>AgeSq</i>	0.000336 (0.000299)		−0.000526 (0.000741)	−0.000563 (0.000537)
<i>Male</i>	0.147 (0.156)		−0.522 (0.375)	−0.256 (0.267)
<i>Education</i>	0.0160 (0.100)	−0.127 (0.249)	−0.304 (0.280)	−0.222 (0.186)
<i>Income</i>	0.184*** (0.0451)	−0.113 (0.128)	−0.226* (0.134)	−0.161* (0.0906)
<i>Constant 1</i>	−0.643 (0.644)			
<i>Constant 2</i>	0.955 (0.650)			
<i>Constant</i>		−3.537*** (0.516)	−4.117*** (1.553)	−3.414*** (1.066)
Observations	314	314	314	314
Fixed effect	Yes	Yes	Yes	Yes
Chi-Square		10.65	17.07	18.31
P-value		0.001	0.000	0.000

Note: Standard errors (robust) in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

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